



Instructor: Date:

Learner: Regional Branch:

Program: Center:

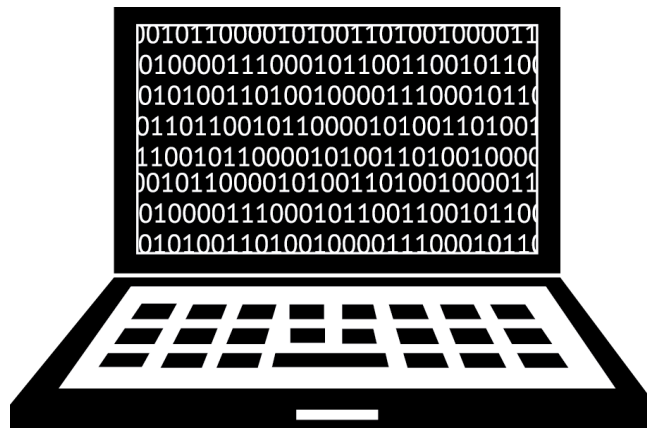
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Let's Start!

Abstraction Data Types



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain what data abstraction is.
- I can identify information related to data abstraction.
- I can ask questions about data abstraction to a group of co-workers.

Ask your partner:

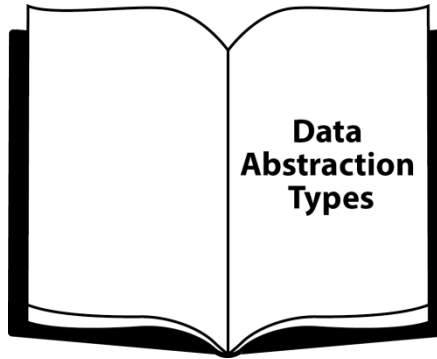
- In what contexts is data management useful? Give examples.
- What do you think the process of abstracting data is about?





Let's Explore!

Welcome to our team!



Instructions: Read the text below.

- Co-worker:** Welcome to our team, Tim. Please open your handbook on page 34, we need to explain to you the significance of data abstraction. Our team works with Abstract data types.
- Tim:** What is it?
- Co-worker:** Abstract data types is a technique to abstract data. To abstract data means to specify, simplify, and encapsulate data, in order to obtain a "generic" type with particular storage features and specific operations to perform on the data type.
- Tim:** Can you please be more specific?
- Co-worker:** To have a complete abstract stack data type, it has to include a structure for sorting data, as a list, an array, or a queue, and a set of operations such as adding or removing integers from the stack.
- Tim:** But, how can we understand data types?
- Co-worker:** To understand data types, we need to identify its advantages. The main advantage of this technique is that it splits structure and functionality from implementation. So you can modify the data type class, or embodiment, without affecting any program.
- Tim:** What do you mean by class?
- Co-worker:** The data class is like the encapsulation of an object, it's the code-template for the creation of objects; it provides the variables, and the behaviors or operations that the programmer decides.
- Tim:** And what do you mean by object?
- Co-worker:** An object is a particular entity of data, to access the data contained in an object, we must have the functions (the class) of the abstract data type. Programmers can change those functions without altering the implementation of the object.
- Tim:** Mention more advantages of the technique, please.
- Co-worker:** Another pro of Abstract Data Types is the inheritance. This means that you can have better derived versions with the printing functions of the original one.

**Let's Practice!**

I do* need to know more about data abstraction.

*We use "do" in this case, to reaffirm something that is extremely necessary

Instructions: Write the words to correctly complete the sentences. Use words from the box below. Read the sentences aloud. Check your pronunciation with your instructor.

To go deeper / To conclude with / To explain

1. _____ this concept, we can say that abstract data is a technique.
2. _____ into this term, we should know that to abstract data means to specify and simplify information.
3. _____ the definition, we want to mention that this technique allows us to obtain a "generic" type of data with particular storage features.

**Let's Practice!**

Can you answer these questions?

Instructions: Organize the mystery sentences. Read the sentences aloud. Check your pronunciation with your instructor.

1. advantages / the / of / abstraction? / What / data / advantages / are

2. is / it / Why / to / integers / a / stack? / remove / from / necessary

3. data / is / an / advantage / in / Why / abstract / types? / inheritance





Extension Activity

Let's define data abstraction!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search information about data abstraction.
- Write at least ten mean ideas about data abstraction.
- Write a summary of data abstraction in two paragraphs.
- Use infinitive of purposes, countable and uncountable nouns.
- Formulate five questions about data abstraction.
- Read your summary aloud and prepare questions to a partner.

Self-Assessment

Yes

No

- I Search information about data abstraction.
- I identify relevant information about data abstraction types.
- I answer questions about data abstraction.

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Target Words

- **integer** (noun): a whole number, as 1,2, 3...
e.g. A set of operations, such as adding or removing integers from the stack.
- **split** (verb): to separate
e.g. The main advantage of this technique is that it splits structure and functionality from implementation.
- **inheritance** (noun): something acquire from a predecessor
e.g. Another pro of Abstract data types is the inheritance.
- **storage** (noun): place you store the data
e.g. In order to obtain a "generic" type of data with particular storage features.
- **stack** (noun): a pile, an amount of something
e.g. A set of operations, such as adding or removing integers from the stack.
- **array** (noun): a layout displayed
e.g. It has to include a structure for sorting data, either as a list or as an array.



Learning Points

Infinitives of purpose to start a sentence

We use infinitives to express the purpose of an action.

The infinitive of verbs always is used to express the goal or purpose of one or more actions. You use the infinitive to start a sentence, by following this structure:

To + base form of the verb.

Example:

To develop a computer game, I need to work hard.

Remember that in this case, you can use the infinitive as the subject of the sentence.

For example:

To design a software, I need to know all the requirements.

However, you can also use infinitives as the complement of a sentence or as the object of the sentence

For example:

*I need to know all the requirements **to** design a software (Complement of a sentence)*

*I need **to** know all the requirements to design a software. (Infinitive as the object of a sentence)*

Infinitives of purpose	
Purpose	Actions
• To analyze data,	<i>we need to know the context.</i>
• To interpret the context,	<i>the engineer has to know the programming language.</i>

Examples

- *To have a complete abstract stack data type, it has to include a structure for sorting data, as a list, an array, or a queue, and a set of operations such as adding or removing integers from the stack.*
- *To understand data types, we need to identify its advantages.*
- *To access the data contained in an object, we must have the functions (the class) of the abstract data type.*



Learning Points

Countable and Uncountable Nouns Related to Software Programming and Data

There are nouns in software programming which are countable or uncountable.

In software programming there are several countable and uncountable nouns.

Countable nouns are things or people we can number as a single unit and/or as plurals.

Examples:

A record- Records.

A structure- Structures.

Remember that when you have more than one item, you use the plural form of the noun; plurals are usually made by adding an "s" to the singular form of the noun.

Examples:

Operation- operations

Representation- representations

Be careful! Not all the plurals are form by adding an "s" or "es", there some irregular plurals.

Let's see some examples:

Singular - Plural

Medium - Media

Phenomenon - Phenomena

Data - Datum

Uncountable nouns are things we cannot conceive as units, we cannot number them.

Le't see some examples:

Nouns in software programming	
Countable	Uncountable
• <i>Representations.</i> • <i>Queue.</i> • <i>Records.</i> • <i>DAQs.</i> • <i>Codes.</i> • <i>Data.</i>	• <i>Software.</i> • <i>Acquisition.</i> • <i>Abstraction.</i> • <i>Encapsulation</i>

Examples

- *It has to include a structure for sorting data, as a list, an array, or a queue.*
- *The data class is like the encapsulation of an object.*

